

For questions 1-2, consider the expression $15y^3 - 10y^2 - 25y$.

1. Write an equivalent expression that uses the greatest common monomial factor.

$$5y(3y^2 - 2y - 5)$$

2. Write an equivalent expression that does not use the greatest common monomial factor.

Answers may vary. Example: $5(3y^3 - 2y^2 - 5y)$

For questions 3-6, write an equivalent expression for the given expression that uses the greatest common monomial factor.

3. $3f^3g^2 + 15fg^3 - 27f^2g^2$

$$3fg^2(f^2 + 5g - 9f)$$

4. $63s^5h^5 + 42s^3h^6 + 35s^4h^2$

$$7s^3h^2(9s^2h^3 + 6h^4 + 5s)$$

5. $15d^3b^3 - 60db^2$

$$15db^2(d^2b - 4)$$

6. $12x^7y^6 - 108x^3y^5 - 84x^6y^4$

$$12x^3y^4(x^4y^2 - 9y - 7x^3)$$

For questions 7-10, find the greatest common monomial factor of the expression.

7. $13a^3 - 39a^2b$

$13a^2$

8. $12q^2c^2 + 30qc + 42q^8c^4$

$6qc$

9. $10r^7m^5 - 55r^2m^3 + 40r^5m$

$5r^2m$

10. $18x^5y^3 - 40x^6y^6 - 38xy$

$2xy$

For questions 1-4, consider the trinomial $x^2 - 5x - 14$.

1. How many $+x$ tiles are needed for an algebra tiles diagram of the trinomial?

2

2. How many $-x$ tiles are needed for an algebra tiles diagram of the trinomial?

-7

3. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 5x - 14$.

 x

-7

[illegible]

4. Complete the table for $x^2 - 5x - 14$.

	x	<u><u>-7</u></u>
x	x^2	<u><u>$-7x$</u></u>
<u><u>$+2$</u></u>	<u><u>$2x$</u></u>	-14

For questions 5-8, consider the trinomial $x^2 + 8x - 9$.

5. How many $+x$ tiles are needed for an algebra tiles diagram of the trinomial?

9

6. How many $-x$ tiles are needed for an algebra tiles diagram of the trinomial?

-1

7. Draw a diagram of the algebra tiles that show the trinomial $x^2 + 8x - 9$.

 x

+ 9

[illegible]

8. Complete the table for $x^2 + 8x - 9$.

	x	<u>$+9$</u>
x	x^2	<u>$9x$</u>
<u>-1</u>	<u>$-x$</u>	-9

9. Complete the table for $x^2 + x - 30$.

	x	<u>$+6$</u>
x	x^2	<u>$6x$</u>
<u>-5</u>	<u>$-5x$</u>	-30

10. Complete the table for $x^2 - 11x + 28$.

	x	<u>-4</u>
x	x^2	<u>$-4x$</u>
<u>-7</u>	<u>$-7x$</u>	28

For questions 1-5, factor each quadratic expression using the method of your choice.

1. $b^2 - 2b - 63$

$(b - 9)(b + 7)$

2. $h^2 + 4h - 77$

$(h + 11)(h - 7)$

3. $x^2 - 5x - 50$

$(x - 10)(x + 5)$

4. $y^2 - 13y + 36$

$(y - 9)(y - 4)$

5. $x^2 - x - 132$

$(x - 12)(x + 11)$

For questions 6-7, consider the trinomial $f^2 - 3f - 10$.

6. Complete the area model.

	f	<u>-5</u>
f	f^2	<u>$-5f$</u>
<u>$+2$</u>	<u>$2f$</u>	-10

7. What are the two binomial factors for $f^2 - 3f - 10$?

$(f - 5) \text{ and } (f + 2)$

For questions 8-9, consider the trinomial $m^2 - 22m + 120$.

8. Complete the area model.

	<u>m</u>	<u>-10</u>
<u>m</u>	<u>m^2</u>	<u>$-10m$</u>
<u>-12</u>	<u>$-12m$</u>	<u>$+120$</u>

9. What are the two binomial factors for $m^2 - 22m + 120$?

$(m - 10)$ and $(m - 12)$

10. Factor $x^2 - x - 20$ using factoring by grouping.

$(x^2 + 4x) + (-5x - 20)$
 $x(x + 4) + (-5)(x + 4)$
 $(x - 5)(x + 4)$

For questions 1-5, factor each quadratic expression using the method of your choice.

1. $5b^2 + 25b + 20$

$$5(b^2 + 5b + 4)$$

$$5(b + 1)(b + 4)$$

2. $4a^2 - 13a - 35$

$$(4a + 7)(a - 5)$$

3. $2x^2 + 5x - 33$

$$(2x + 11)(x - 3)$$

4. $3k^2 - 27k + 60$

$$(3k - 15)(k - 4)$$

$$3(k - 5)(k - 4)$$

5. $2z^2 + 13z - 70$

$$(2z - 7)(z + 10)$$

For questions 6-7, consider the trinomial $3x^2 - 2x - 5$.

6. Complete the area model.

	$3x$	<u>-5</u>
x	$3x^2$	<u>$-5x$</u>
<u>$+1$</u>	<u>$3x$</u>	-5

7. What are the two binomial factors for $3x^2 - 2x - 5$?

$(3x - 5)$ and $(x + 1)$

For questions 8-9, consider the trinomial $2m^2 + 11m - 63$.

8. Complete the area model.

	<u>$2m$</u>	<u>-7</u>
<u>m</u>	<u>$2m^2$</u>	<u>$-7m$</u>
<u>$+9$</u>	<u>$18m$</u>	<u>-63</u>

9. What are the two binomial factors for $2m^2 + 11m - 63$?

$(2m - 7)$ and $(m + 9)$

10. Factor $3x^2 + 22x + 24$ using factoring by grouping.

$(3x^2 + 18x) + (4x + 24)$
 $3x(x + 6) + 4(x + 6)$
 $(3x + 4)(x + 6)$

For questions 1-3, factor each quadratic expression using the method of your choice.

1. $4b^2 + 8b - 45$

$(2b - 5)(2b + 9)$

2. $12x^2 + 7x - 49$

$(3x + 7)(4x - 7)$

3. $10d^2 - 9d + 2$

$(2d - 1)(5d - 2)$

For questions 4-5, consider the trinomial $9x^2 - 27x + 14$.

4. Write an expanded form of the trinomial that can be used to factor the expression by grouping.

$(9x^2 - 6x) + (-21x + 14)$

5. Factor the expression by grouping.

$3x(3x - 2) - 7(3x - 2)$
 $(3x - 7)(3x - 2)$

For questions 6-7, consider the trinomial $6x^2 - 7x - 20$.

6. Complete the area model.

	$3x$	<u>$+4$</u>
$2x$	$6x^2$	<u>$8x$</u>
<u>-5</u>	<u>$-15x$</u>	-20

7. What are the two binomial factors for $6x^2 - 7x - 20$?

$(3x + 4)$ and $(2x - 5)$

For questions 8-9, consider the trinomial $8t^2 + 18t - 5$.

8. Complete the area model.

	<u>$4t$</u>	<u>-1</u>
<u>$2t$</u>	<u>$8t^2$</u>	<u>$-2t$</u>
<u>$+5$</u>	<u>$20t$</u>	<u>-5</u>

9. What are the two binomial factors for $8t^2 + 18t - 5$?

$(4t - 1)$ and $(2t + 5)$

10. Factor $10x^2 - 3x - 1$ using factoring by grouping.

$(10x^2 - 5x) + (2x - 1)$
 $5x(2x - 1) + 1(2x - 1)$
 $(5x + 1)(2x - 1)$

For questions 1-6, factor each expression by grouping.

1. $2y^4 - 6y^3 + 14y^2 - 42y$

$$\begin{aligned} &y(2y^3 - 6y^2 + 14y - 42) \\ &y[(2y^3 - 6y^2) + (14y - 42)] \\ &y[2y^2(y - 3) + 14(y - 3)] \\ &y(2y^2 + 14)(y - 3) \\ &2y(y^2 + 7)(y - 3) \end{aligned}$$

2. $15x^2 + 34x + 15$

$$\begin{aligned} &(15x^2 + 9x) + (25x + 15) \\ &3x(5x + 3) + 5(5x + 3) \\ &(3x + 5)(5x + 3) \end{aligned}$$

3. $-6a + 24ab - 20b + 5$

$$\begin{aligned} &(-6a + 24ab) + (-20b + 5) \\ &-6a(1 - 4b) + 5(-4b + 1) \\ &(-6a + 5)(-4b + 1) \end{aligned}$$

4. $6x^2 - x - 40$

$$\begin{aligned} &(6x^2 - 16x) + (15x - 40) \\ &2x(3x - 8) + 5(3x - 8) \\ &(2x + 5)(3x - 8) \end{aligned}$$

5. $3x^2 + 5x + 6x + 10$

$$\begin{aligned} &(3x^2 + 5x) + (6x + 10) \\ &x(3x + 5) + 2(3x + 5) \\ &(x + 2)(3x + 5) \end{aligned}$$

6. $5x^2 + 52x + 63$

$$\begin{aligned} &(5x^2 + 45x) + (7x + 63) \\ &5x(x + 9) + 7(x + 9) \\ &(5x + 7)(x + 9) \end{aligned}$$

For questions 7-10, consider the trinomial $16x^2 + 16x - 5$.

7. List all the factor pairs for -80 .

$-80, 1$ $-40, 2$ $-20, 4$ $-16, 5$ $-10, 8$
 $80, -1$ $40, -2$ $20, -4$ $16, -5$ $10, -8$

8. Find the factor pair for -80 that add to the sum of 16.

$20, -4$

9. Use question 8 to write the expanded form of $16x^2 + 16x - 5$ that can be used to factor the expression by grouping.

$(16x^2 - 4x) + (20x - 5)$

10. Factor the expression $16x^2 + 16x - 5$ by grouping.

$4x(4x - 1) + 5(4x - 1)$
 $(4x + 5)(4x - 1)$

For questions 1-2, consider the perfect square trinomial $9x^2 + 24x + 16$.

1. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

$$(3x)^2 + 2(3x)(4) + 4^2$$

2. Factor the perfect square trinomial.

$$(3x + 4)^2$$

For questions 3-4, consider the perfect square trinomial $16x^2 - 40x + 25$.

3. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

$$(4x)^2 - 2(4x)(5) + 5^2$$

4. Factor the perfect square trinomial.

$$(4x - 5)^2$$

For questions 5-8, determine whether the quadratic expression is a perfect square trinomial. Then, factor the expression.

5. $w^2 - 20w - 100$

No.
Cannot be factored

6. $f^2 - 2fg + g^2$

Yes.
 $(f - g)^2$

7. $25v^2 - 9$

No.
 $(5v + 3)(5v - 3)$

8. $9y^2 - 81y + 18$

No.
 $9(y^2 - 9y + 2)$

For questions 9-10, factor the trinomial.

9. $25a^2 + 120a + 144$

$(5a + 12)^2$

10. $p^2 - 22p + 121$

$(p - 11)^2$

Algebra 1

Unit 7

Lesson 8 Homework

Name Answer key

Date _____

Period _____

For questions 1-2, consider the expression $16m^2 - 100p^2$.

1. Rewrite the expression as a difference of two squares.

$$(4m)^2 - (10p)^2$$

2. Factor the expression.

$$(4m + 10p)(4m - 10p)$$

For questions 3-4, consider the expression $144a^2 - 49b^2$.

3. Rewrite the expression as a difference of two squares.

$$(12a)^2 - (7b)^2$$

4. Factor the expression.

$$(12a + 7b)(12a - 7b)$$

For questions 5-8, determine whether the quadratic expression is a difference of two squares. Then, factor the expression.

5. $64d^2 - 121g^2$

Yes.

$(8d + 11g)(8d - 11g)$

6. $9f^2 + 48f + 64$

Not the difference of two squares.

$(3f + 8)^2$

7. $1 - r^2$

Yes.

$(1 + r)(1 - r)$

8. $-121 + t^2$

Yes.

$(t + 11)(t - 11)$

For questions 9-10, factor the expression.

9. $x^2 - 2x + 1$

$(x - 1)^2$

10. $16y^2 - 25x^2$

$(4y + 5x)(4y - 5x)$